



HOCHSCHULE
HAMM-LIPPSTADT

Autonomous Prototype Car Project

(Team D5)

Team members:

Ali Abdelkhalek
Mohamed Awis
Yassin Elhalawany

Supervisors:

Prof/ Dr Stefan Henkler
Prof/ Dr Faezeh Pasandideh

Project Overview

From system engineering to real-time line following and obstacle avoidance

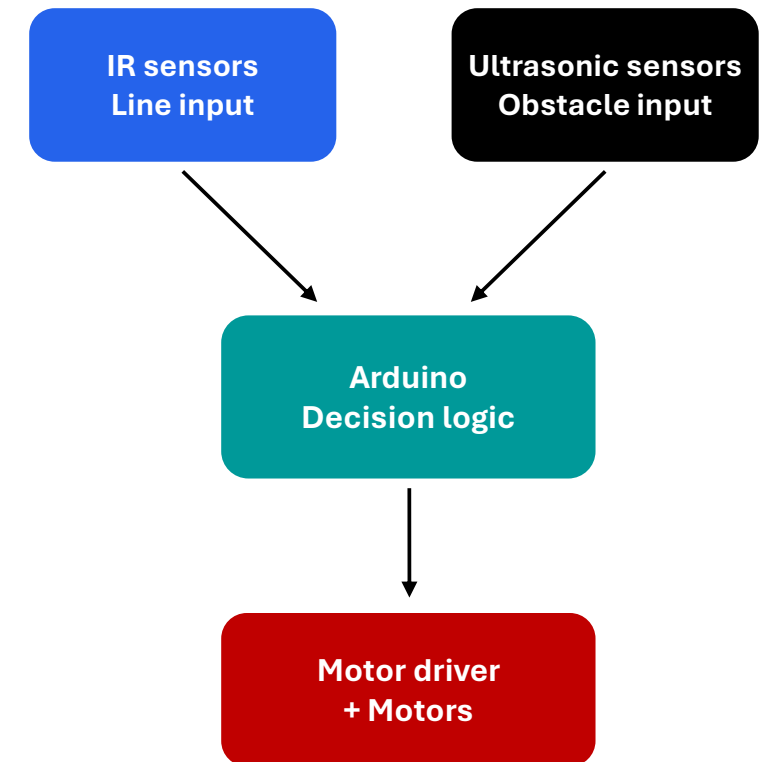
Project goal

- Follow a path autonomously using IR sensor feedback.
- Detect obstacles early using an ultrasonic sensor.
- Avoid the obstacle, recover the line, and continue without manual control.

Engineering path



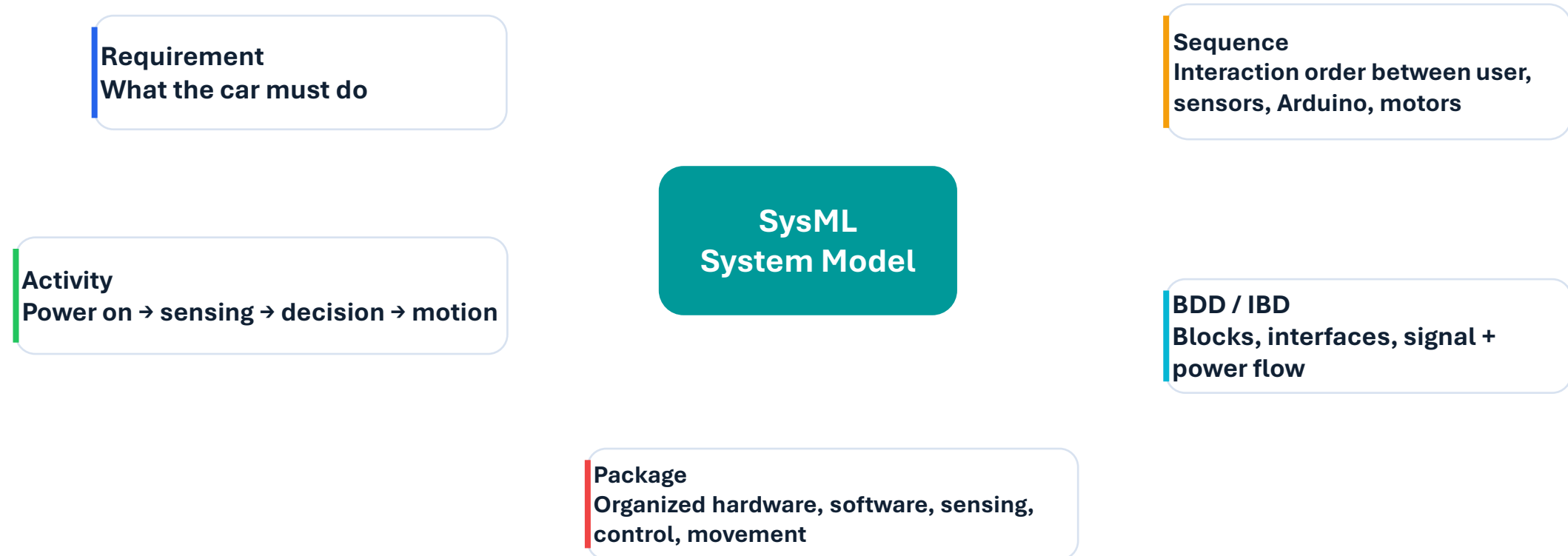
Sense → Decide → Move



Core autonomous loop

System Engineering Using SysML

We modeled the prototype before implementation so the behavior, structure, and interfaces were clear.

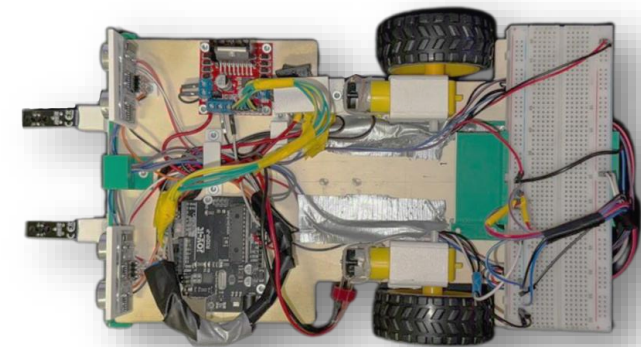
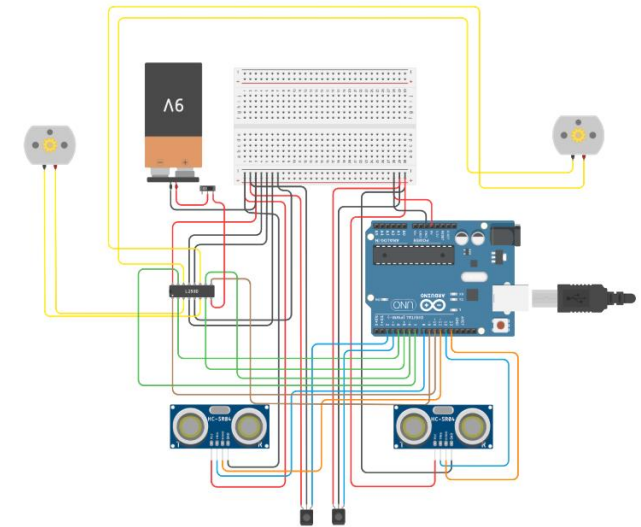
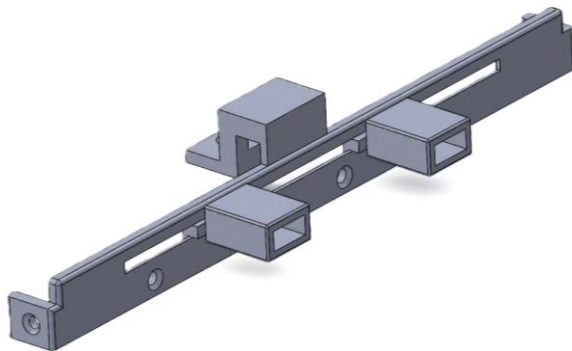


SysML connected the idea to the real implementation and reduced trial-and-error during integration.

TinkerCAD Simulation & 3D Design

After SysML, we verified the circuit in TinkerCAD, then designed and assembled the physical layout.

- Created a TinkerCAD circuit simulation with Arduino, breadboard, microcontroller, sensors, battery wiring, motor control driver and motors.
- Checked signal and power connections before assembling the real prototype.
- Designed and 3D printed the structure to hold all main components in fixed positions.
- Ensured neat and organized cable management.
- Created an adjustable IR sensor holder to tune width during testing.



Milestone 1: Line Following

We adapted the software concept to match the real sensor spacing of our 3D design.

Initial concept

Move forward when IR sensors directly detect the black line.

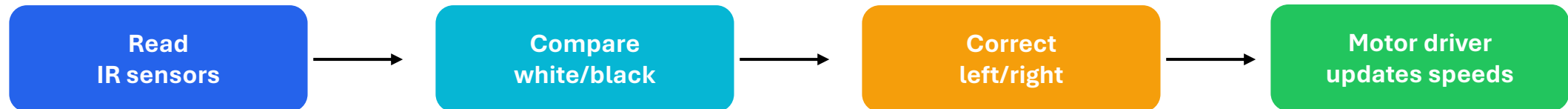
Evaluation issue

IR sensors could not be placed very close together; direct black-line detection was not reliable.

Final concept

Use white-surface readings to adjust the car around the black line.

Line-following control loop



Result

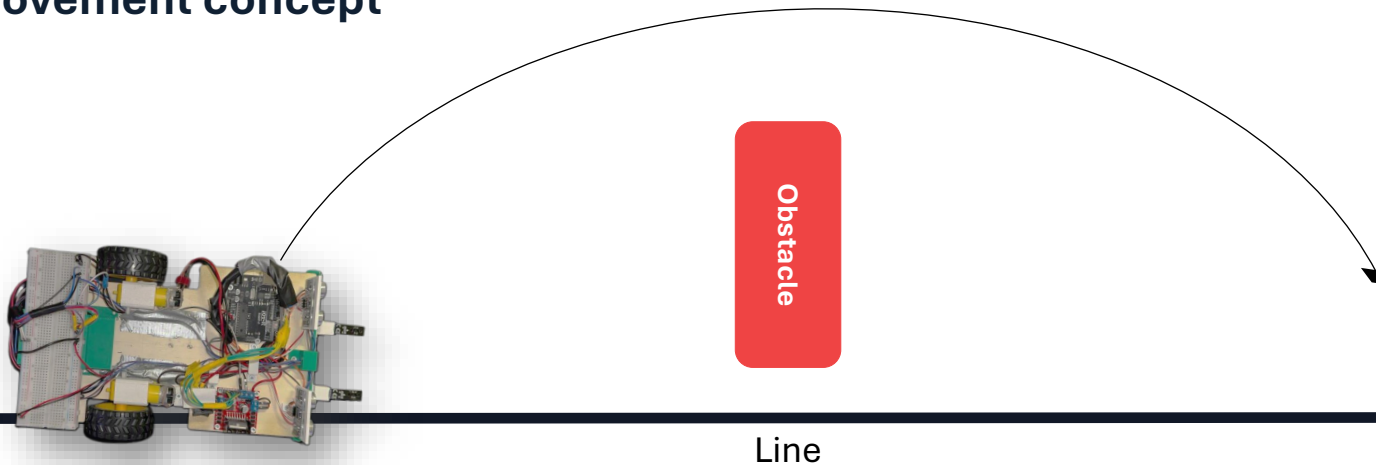
The final logic matched our mechanical design better and made line following more effective with higher speeds recording fastest lap in our group

Milestone 2: Obstacle Avoidance

We added an avoidance behavior that temporarily takes priority over normal line following.



Avoidance movement concept



Avoidance is not only passing the obstacle, but it must also recover the line successfully.

Conclusion: What We Achieved and Learned

A complete prototype was built through design, modeling, implementation, and testing.

Achieved

Integrated 3D design, sensors, Arduino logic, motor driver, and movement into one autonomous prototype.

Main learning

Mechanical design, component placement, code logic and speed optimization.

Evaluation value

Testing changed our line-following concept and improved the obstacle-avoidance timing.

PID control
Smoother correction

More IR sensors
Better line recovery

Future improvements

Enhanced motor control
driver

Specified PCB
for the project



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Thank you for your attention

We are happy to answer your questions!